

Chairperson: Voisin, Kevin

Secretary: Voisin, Kevin

Present:

Carrino, Francesco

Voisin, Kevin

Schedule

1	Discussion Points & Methodological Reflection	1
1.1	High-Frequency Torque Availability (40 kHz)	1
1.2	Report Status & Statistical Analysis	1
2	Data Engineering & Modeling Strategy	1
2.1	Feature Extraction & Dimensionality Reduction	1
3	Action Items	2

Start of meeting

1 Discussion Points & Methodological Reflection

1.1 High-Frequency Torque Availability (40 kHz)

- **Core Issue:** The current Temporal Fusion Transformer (TFT) model risks being biased or misconfigured because it attempts to predict a single aggregated torque value (Y) using highly dynamic, high-frequency sensor signals (X). Different vibration signatures could map to the same target value, introducing noise.
- **Proposed Solutions:**
 - Contact Rémy Borgeat to verify if it is technically possible to extract the torque data at the native sensor frequency (40kHz, i.e., every $\frac{1}{40000}$ s).
 - As a temporary workaround, aggregate and summarize the data at the pass level (sensor_file level) to train a lighter baseline model (e.g., LSTM).

1.2 Report Status & Statistical Analysis

- **Status Update:** The drafting of the Bachelor thesis report is slightly delayed.
- **Action Item:** Focus on defining a robust exploratory data analysis (EDA) and visualization strategy. The goal is to mathematically and visually demonstrate that the raw vibration signals contain meaningful physical information correlated with the process.

2 Data Engineering & Modeling Strategy

2.1 Feature Extraction & Dimensionality Reduction

- **Feature Pipeline:** Implement a comprehensive feature extraction process using the **TSFEL** library, followed by a feature selection step utilizing **TSFRESH** to retain only the most predictive variables.

- **Advanced Features:** Integrate the **Hurst Exponent** into the feature pool to capture long-term memory and stationary properties of the time series.
- **Alternative Layouts:** Test a time-slicing strategy by sequencing the passes into three distinct temporal phases ($\frac{1}{3}$, $\frac{1}{3}$, $\frac{1}{3}$) or other split, to capture the evolution of the signals.
- 25 • **Visualization:** Substitute Principal Component Analysis (PCA) with **t-SNE** (t-Distributed Stochastic Neighbor Embedding) to improve the non-linear visualization and clustering of the high-dimensional feature space.

3 Action Items

	Actions
30	Contact Rémy Borgeat regarding the availability of high-frequency torque data.
	Implement the data aggregation at the pass level and evaluate a lightweight LSTM baseline.
	Develop the t-SNE visualization and incorporate TSFEL/TSFRESH features with Hurst Exponent.
	Catch up on the thesis report writing and draft the initial statistical analysis section.

35 11:40 AM Meeting Adjourned